MFML AND MFDS

MFML and MFDS Team
WILPD - CSIS

BITS Pilani

BITS Pilani

* ZC416 Mathematical Foundations for Machine Learning / Data Science

Topic: # 5 Singular Value Decomposition

INTRODUCTION

- In the previous lecture, we discussed eigenvalues and eigenvectors of matrices
- In this lecture, we will look at two related methods for factorizing matrices into canonical forms.
- The first one is known as Eigenvalue decompostion. It uses the concepts of eigenvalues and eigenvectors to generate the decomposition
- The second method known as singular value decomposition or SVD is applicable to all matrices.

DIAGONAL MATRICES

- A diagonal matrix is a matrix that has value zero on all off diagonal elements.

$$D = \begin{bmatrix} d_1 & & \\ & \ddots & \\ & & d_n \end{bmatrix}$$

- For a diagonal matrix D, the determinant is the product of its diagonal entries.
- A matrix power D^k is given by each diagonal element raised to the power k.
- Inverse of a diagonal matrix is obtained by taking inverse of non-zero diagonal entry.

DIAGONALIZABLE MATRICES

- A matrix $A \in \mathbb{R}^{n \times n}$ is diagonalizable if there exists an invertible matrix $P \in \mathbb{R}^{n \times n}$ and a diagonal matrix D such that $D = P^{-1}AP$
- In the definition of diagonalization, it is required that **P** is an invertible matrix. Assume $p_1, p_2, \ldots.., p_n$ are the n columns of **P**
- Rewriting we get $AP = PD$. By observing that D is a diagonal matrix, we can simplify as

$$Ap_i = \lambda_i p_i$$

where λ_i is the i^{th} diagonal entry in D.

Diagonalizable Matrix

- Consider a square matrix

$$A = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$$

- Consider the invertible matrix

$$\mathbf{P} = \begin{bmatrix} -2 & 1 \\ 1 & 1 \end{bmatrix}$$

- Now consider the product $\mathbf{P}^{-1}\mathbf{AP}$ as follows

$$\begin{bmatrix} -2 & 1 \\ 1 & 1 \end{bmatrix}^{-1} \cdot \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix} \cdot \begin{bmatrix} -2 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & 5 \end{bmatrix}$$

EIGENDECOMPOSITION OF A MATRIX

- Recall the existence of eigenvalues and eigenvectors for square matrices
- Eigenvalues can be used to create a matrix decomposition known as Eigenvalue Decomposition
- A square matrix $A \in \mathbb{R}^{n \times n}$ can be factored into

$$A = PDP^{-1}$$

- where P is an invertible matrix of eigenvectors of A assuming we can find n eigenvectors that form a basis of $\mathbb{R}^n$
- and D is a diagonal matrix whose diagonal entries are the eigenvalues of A

EXAMPLE OF EIGENDECOMPOSITION

Let us compute the eigendecomposition of the matrix A

$$\mathbf{A} = \begin{bmatrix} 2.5 & -1 \\ -1 & 2.5 \end{bmatrix}$$

- Step 1: Find the eigenvalues and eigenvectors

$$\mathbf{A} - \lambda \mathbf{I} = \begin{bmatrix} 2.5 - \lambda & -1 \\ -1 & 2.5 - \lambda \end{bmatrix}$$

- The characteristic equation is given by $\det(\mathbf{A} - \lambda \mathbf{I}) = \mathbf{0}$
- This leads to the equation $\lambda^2 - 5\lambda + \frac{21}{4} = 0$
- Solving the quadratic equation gives us $\lambda_1 = 3.5$ and $\lambda_2 = 1.5$

EXAMPLE OF EIGENDECOMPOSITION

- The eigenvector corresponing to $\lambda_1 = 3.5$ is derived as
$$p_1 = \begin{bmatrix} -\frac{1}{\sqrt{2})} \\ \frac{1}{\sqrt{2}} \end{bmatrix}$$
- The eigenvector corresponing to $\lambda_1 = 1.5$ is derived as
$$p_2 = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}$$
- Step 2 : Construct the matrix **P** to diagonalize **A**

$$\mathbf{P} = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

Example of Eigendecomposition

- The inverse of matrix P is given by

$$\mathbf{P}^{-1} = \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

- The eigendecompostion of the matrix A is given by

$$\mathbf{A} = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} 3.5 & 0 \\ 0 & 1.5 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

- In summary we have obtained the required matrix factorization using eigenvalues and eigenvectors.

SYMMETRIC MATRICES AND DIAGONALIZABILITY

- Recall that a matrix A is called symmetric matrix if $\mathbf{A} = \mathbf{A}^\mathsf{T}$

$$\mathbf{A} = \begin{bmatrix} -2 & 1 \\ 1 & 1 \end{bmatrix}$$

- A Symmetric matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ can always be diagonalized.
- This follows directly from the spectral theorem discussed in the previous lecture
- Moreover, the spectral theorem states that we can find an orthogonal matrix $\mathbf{P}$ of eigenvectors of $\mathbf{A}$.

Motivation for Singular Value Decomposition

- The singular value decomposition or (SVD) of a matrix is a central matrix decomposition method in linear algebra.
- The eigenvalue decomposition is applicable to square matrices only.
- The singular value decomposition exists for all rectangular matrices
- SVD involves writing a matrix as a product of three matrices $\boldsymbol{U}$, Σ and $\boldsymbol{V}^T$.
- The three component matrices are derived by applying eigenvalue decomposition discussed previously

SINGULAR VALUE DECOMPOSITION THEOREM

- Let $\mathbf{A} \in \mathbb{R}^{\mathbf{m \times n}}$ be a rectangular matrix. Assume that A has rank r.
- The Singular value decomposition of $\mathbf{A}$ is defined as $\mathbf{A} = U \Sigma V^T$
- $\mathbf{U} \in \mathbb{R}^{m \times m}$ is an orthogonal matrix with column vectors u_i where $i = 1, ..., m$
- $\mathbf{V} \in \mathbb{R}^{n \times n}$ is an orthogonal matrix with column vectors v_j where $j = 1, ..., n$
- Σ is an m × n matrix with $\Sigma_{ii} = \sigma_i > 0$
- The diagonal entries Σ_i , $i = 1, ..., r$ of Σ are called the singular values.
- By convention, the singular values are ordered i.e. $\Sigma_{ii} > \Sigma_{jj}$ for $i < j$.

PROPERTIES OF Σ

- The singular value matrix Σ is unique.
- Observe that the $\Sigma \in \mathbb{R}^{m \times n}$matrix is rectangular. In particular, Σ is of the same size as **A**.
- This means that Σ has a diagonal submatrix that contains the singular values and needs additional zero padding.
- Specifically, if $m > n$, then the matrix Σ has diagonal structure up to row n and then consists of zero rows.
- If $m < n$, the matrix Σ has a diagonal structure up to column m and columns that consist of 0 from $m + 1$ to n.

CONSTRUCTION OF V

- It can be observed that

$$\boldsymbol{A^T A = V \Sigma^T \Sigma V^T}$$

- Since $A^T A$ has the following eigendecomposition

$$\mathbf{A^T A = PDP^T}$$

- Therefore, the eigenvectors of $A^T A$ that compose P are the right-singular vectors V of A.
- The eigenvalues of $A^T A$ are the squared singular values of Σ

CONSTRUCTION OF U

- It can be observed that

$$AA^T = U\Sigma V^T V\Sigma^T U^T$$

- Since $\mathbf{AA^T}$ has the following eigendecomposition

$$AA^T = SDS^T$$

- Therefore, the eigenvectors of $\mathbf{AA^T}$ that compose **S** are the left-singular vectors **U** of **A**

CONSTRUCTION OF U CONTINUED

- $\mathbf{A} = \boldsymbol{U}\Sigma\boldsymbol{V}^{\boldsymbol{T}}$ can be rearranged to obtain a simple formulation for u_i
- By postmultiplying by **V** we get $\boldsymbol{AV} = \boldsymbol{U}\Sigma\boldsymbol{V}^{\boldsymbol{T}}\boldsymbol{V}$
- By observing that **V** is orthogonal we obtain a simple form

$$\boldsymbol{AV} = \boldsymbol{U}\Sigma$$

- This is equivalent to the following

$$u_i = \frac{1}{\sigma_i}\mathbf{A}v_i \quad \forall i = 1, 2, ..., r$$

Computing Singular Value Decomposition 1

- We want to find SVD of the following rectangular matrix $\mathbf{A}$

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 1 \\ -2 & 1 & 0 \end{bmatrix}$$

- Let us consider the matrix $\mathbf{A}^T\mathbf{A}$ derived from $\mathbf{A}$ given by

$$\mathbf{A}^T\mathbf{A} = \begin{bmatrix} 5 & -2 & 1 \\ -2 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

- It is a symmetric matrix

Computing Singular Value Decomposition 2

- Derive the eigendecomposition of $\mathbf{A}^T\mathbf{A}$ in the form PDP^T
- The matrix $\mathbf{P}$ is given by

$$\mathbf{P} = \begin{bmatrix} \frac{5}{\sqrt{30}} & 0 & \frac{-1}{\sqrt{6}} \\ \frac{-2}{\sqrt{30}} & \frac{1}{\sqrt{5}} & \frac{-2}{\sqrt{6}} \\ \frac{1}{\sqrt{30}} & \frac{2}{\sqrt{5}} & \frac{1}{\sqrt{6}} \end{bmatrix}$$

- The matrix $\mathbf{D}$ is given by

$$\mathbf{D} = \begin{bmatrix} 6 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Computing Singular Value Decomposition 3

Now we construct the singular value matrix Σ

- The matrix Σ has the dimension same as $\mathbf{A}$. In this case Σ is hence a 2×3 matrix.
- The diagonal entries of submatrix is obtained by taking square root of 6 and 1 respectively.
- Singular-value matrix Σ is given by:

$$\Sigma = \begin{bmatrix} \sqrt{6} & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

- The last column is a column of zeros only

Computing Singular Value Decomposition 4

Left singular vectors as the normalized image of the right singular vectors. Recall that $u_i = \frac{1}{\sigma_i}\mathbf{A}v_i$

- The first vector

$$u_1 = \frac{1}{\sigma_1}\mathbf{A}v_1 = \begin{bmatrix} \frac{1}{\sqrt{5}} \\ \frac{-2}{\sqrt{5}} \end{bmatrix}$$

- The second vector

$$u_2 = \frac{1}{\sigma_2}\mathbf{A}v_2 = \begin{bmatrix} \frac{2}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} \end{bmatrix}$$

Final Step : Combining U, Σ and V

We compile all the three matrices together to generate the SVD

- $$\mathbf{A} = \begin{bmatrix} \frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}} \\ \frac{-2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \end{bmatrix} \begin{bmatrix} \sqrt{6} & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} \frac{5}{\sqrt{30}} & 0 & \frac{-1}{\sqrt{6}} \\ \frac{-2}{\sqrt{30}} & \frac{1}{\sqrt{5}} & \frac{-2}{\sqrt{6}} \\ \frac{1}{\sqrt{30}} & \frac{2}{\sqrt{5}} & \frac{1}{\sqrt{6}} \end{bmatrix}^T$$
- The matrix **U** is an 2×2 matrix satifying orthogonality property.
- The matrix **V** is an 3×3 matrix satifying orthogonality property.

Comparing SVD and EVD

- The left-singular vectors of $\mathbf{A}$ are eigenvectors of $\mathbf{A}\mathbf{A}^{\mathbf{T}}$
- The right-singular vectors of $\mathbf{A}$ are eigenvectors of $\mathbf{A}^T\mathbf{A}$
- The non-zero singular values of $\mathbf{A}$ are the square roots of the nonzero eigenvalues of $\mathbf{A}^T\mathbf{A}$.
- The SVD always exists for any matrix in $\mathbb{R}^{m \times n}$
- The eigendecomposition is only defined for square matrices in $\mathbb{R}^{n \times n}$ and only exists if we can find a basis of eigenvectors of $\mathbb{R}^n$

COMPARING SVD AND EVD

- The vectors in the eigendecomposition matrix **P** are not necessarily orthogonal.
- On the other hand, the vectors in the matrices **U** and **V** in the SVD are orthonormal.
- Both the eigendecomposition and the SVD are compositions of three linear mappings:
- A key difference between the eigendecomposition and the SVD is that in the SVD, domain and codomain can be of different dimensions
- In the SVD, the left and right singular vector matrices **P** and **P** are generally not inverse of each other.

COMPARING SVD AND EVD 3

- In the eigendecomposition, the matrices in decomposition are inverse of each other
- In the SVD, the entries in the diagonal matrix Σ are all real and nonnegative,
- In eigendecomposition diagonal matrix entries need not be real always.
- The leftsingular vectors of $\mathbf{A}$ are eigenvectors of $\mathbf{A}\mathbf{A}^T$
- The rightsingular vectors of $\mathbf{A}$ are eigenvectors of $\mathbf{A}^T\mathbf{A}$.

MATRIX APPROXIMATION

- We considered the SVD as a way to factorize $\mathbf{A} = \boldsymbol{U}\Sigma\boldsymbol{V}^{\boldsymbol{T}}$ into the product of three matrices, where **U** and **V** are orthogonal and Σ contains the singular values on its main diagonal.
- Instead of doing the full SVD factorization, we will now investigate how the SVD allows us to represent a matrix **A** as a sum of simpler matrices $\mathbf{A}_i$
- This representation which lends itself to a matrix approximation scheme that is cheaper to compute than the full SVD.

MATRIX APPROXIMATION CONTINUED

- A matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$ of rank r can be written as a sum of rank-1 matrices so that $\mathbf{A} = \sum_{i=1}^{r} \sigma_i u_i v_i^T$
- The diagonal structure of the singular value matrix Σ multiplies only matching left and right singular vectors $u_i v_i^T$ and scales them by the corresponding singular value σ_i.
- All terms $\sigma_i u_i v_j^T$ vanish for $i \neq j$ because Σ is a diagonal matrix.
- Any term for $i > r$ would vanish because the corresponding singular value is 0.

Rank k Approximation

- We summed up the r individual rank-1 matrices to obtain a rank r matrix **A**.
- If the sum does not run over all matrices $A_i \quad i = 1, \ldots, r$ but only up to an intermediate value k we obtain a rank-k approximation
- The approximation represented by $\hat{\mathbf{A}}(k)$ is defined as follows

$$\hat{\mathbf{A}}(k) = \sum_{i=1}^{k} \sigma_i u_i v_i^T$$

- To measure the difference between **A** and its rank-k approximation we need the notion of a norm which is introduced next

Spectral Norm of a Matrix

- We introduce the notation of a subscript in the matrix norm
- Spectral Norm of a Matrix. For $x \in \mathbb{R}^n$, $x \neq \mathbf{0}$, the spectral norm norm of a matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$ is defined as

$$||\mathbf{A}||_2 = \max_x \frac{||\mathbf{A}x||_2}{||x||_2}$$

  where $||y||_2$ is the euclidean norm of y
- Theorem : The spectral norm of a matrix $\mathbf{A}$ is its largest singular value

Example : Spectral Norm of a Matrix

- Example : Consider the following matrix **A**

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

- Singular value decomposition of this matrix will provide the matrix Σ as follows

$$\Sigma = \begin{bmatrix} 5.465 & 0 \\ 0 & 0.366 \end{bmatrix}$$

- The 2 singular values are 5.4650 and 0.366.
- By definition the spectral norm is the largest singular value.
- Hence, the spectral norm is 5.4650

innovate achieve lead

BITS Pilani

Thank you